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Unlocking Coal Seam Fractures: Enhancing Structural Analysis of Cleats with Horizontal Borehole Image Logs

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Abstract

Cleat identification is crucial for Coal Bed Methane (CBM) production and present-day stress regimes. Cleats which are fractures present in coal seams, typically classified as face and butt cleats, perpendicular to each other and to the coal bedding. The straight-line visualization of cleats in vertical boreholes presents challenges for accurate interpretation. Therefore, the optimal approach is to characterize cleats in horizontal boreholes, allowing more accurate analysis of cleats interpretation.

In this study, an 8-pad, 2.4" Slim Micro Resistivity Imager Tool (SMRIT) was deployed in 10 horizontal Water-Based Mud (WBM) boreholes to acquire high-resolution borehole images. Along with the microresistivity data, petrophysical logs such as gamma ray, resistivity, density and caliper logs were integrated to provide a comprehensive understanding of the subsurface. Data processing was performed, including corrections for magnetic declination and speed correction to ensure accurate image representation. These corrections were crucial for generating both static and dynamic 360° resistivity-based borehole images. Cleats, identified based on their high dip angles (>75°), were manually delineated from dynamic images.

The study processed data from 10 coal seam horizontal boreholes, utilizing dynamic images to clearly distinguish and verify cleats along coal seams. Cleats and other fractures were differentiated by dip angle, as formation fractures (either conductive, resistive, or mixed) appeared similar. In relationship between cleat density and cleat spacing, most of the boreholes show high density of cleats and low cleat spacing. This characterization of cleats significantly improves the effective permeability of otherwise low-permeability coal and improves CBM production. All boreholes showed two distinct cleat orientations (major and minor directions), with an average of 810 cleats per average of 500 m section. Horizontal boreholes provided clearer identification of cleat features compared to vertical boreholes, where orientations were more scattered. The dominant face cleat aligned with the maximum horizontal stress (σ_{Hmax}) in a NW-SE direction, while the subordinate butt cleat, oriented perpendicular to the face cleat, aligned with the minimum stress (σ_{Hmin}) in a NE-SW direction.

The SMRIT proved to be a valuable tool for identifying geological structures in horizontal boreholes, especially cleats within coal seams. In addition to structural interpretation, the tool enabled statistical analysis of cleats and reveals the CBM production and horizontal stresses of coals in the study area. This

approach supports multiscale studies of cleat systems, enhancing the understanding of coal seam fracture characteristics and field stress regimes.

Background

The term "cleat" is commonly used by geologists, engineers and miners since 1925 for fracture identification in coal due to coalification process and basin regional tectonics (Rodrigues et al. 2014). In a simple definition, cleats were described by Abrar et al. (2025) as vertical to nearly vertical fractures present in coal seams. The cleat system in coal seams is theoretically characterized by two sets of sub-parallel fractures, both mostly orthogonal to bedding. These are made up of face cleats which are dominant, with individual surfaces almost planar, persistent, laterally extensive, and widely spaced, and butt cleats which are poorly defined set of natural fractures, orthogonal or nearly orthogonal to face cleats (Fig. 1).

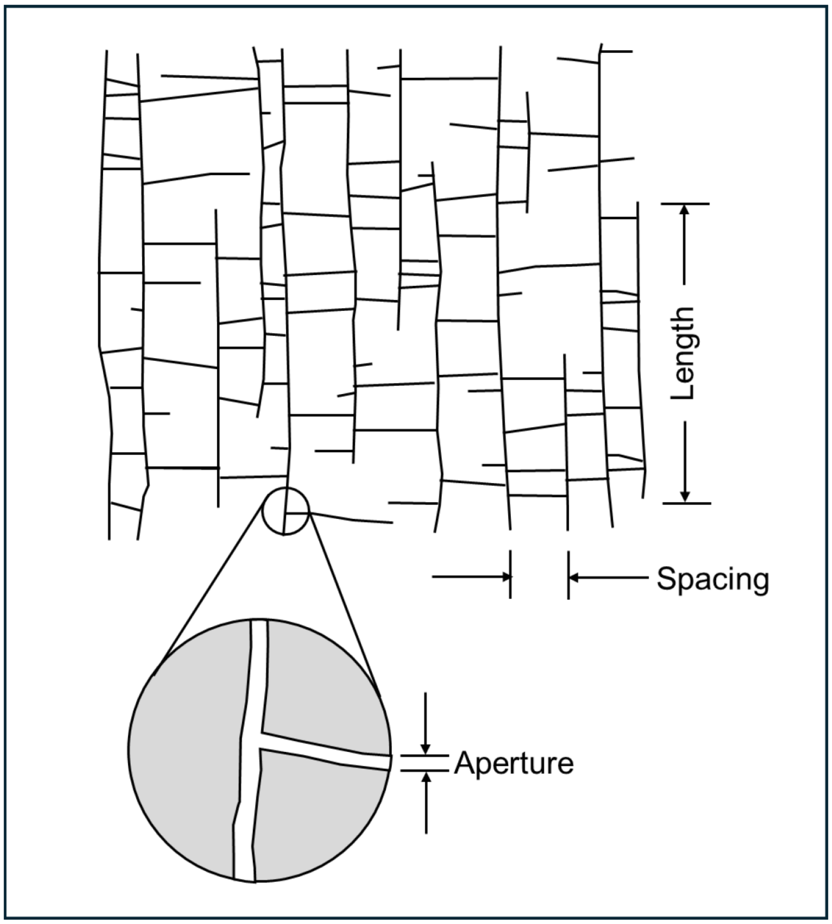

Figure 1—Illustration of cleats in coal seams. Cleat length is parallel to cleat surface and parallel to bedding. Cleat spacing is a distance between two cleats of same set at right angles to cleat surface. Aperture is a perpendicular dimension to fracture surface. Modified after Laubach and Tremain, 1991).

These fracture sets significantly influence fluid flow and mechanical behavior of coal seams (Laubach et al. 1998). With combination of core samples and image logs, cleat system defines the permeability of the coal seam and governs gas production rates in Coal Bed Methane (CBM) reservoirs (Close, 1993). Understanding how cleats are arranged and how they affect fluid flow is the key to improving CBM recovery, ensuring safety and planning reservoir development.

From a geological view, cleats can show how the coal has been stressed and fractured over time. Laubach et al. (1998) also emphasized systematic cleat orientations often align with regional maximum horizontal stress and can be used to interpret present-day stress regimes. The orientation of face cleats tends to parallel the direction of the current or past maximum horizontal stress (σ_{Hmax}). This is because face cleats usually develop perpendicular to the minimum horizontal stress (σ_{Hmin}), aligning with the principal stress direction. Understanding present-day stress helps in wellbore stability analysis, hydraulic fracturing design, fracture permeability prediction and reservoir stimulation and production planning.

Limitation of Cleat Identification in Vertical Boreholes

The orientation of a borehole, whether vertical or horizontal, significantly influences the way cleats (natural fractures in coal) are intersected and interpreted. This, in turn, impacts assessments of gas production, reservoir permeability, drilling strategies, anisotropy evaluation and other critical studies. In vertical boreholes, cleats can be identified based on straight line features in the coal section (Fig. 2). However, there is uncertainty regarding cleat picking because these steeply dipping cleats intersect the borehole wall at high angles, often nearly parallel to the borehole axis. This geometry causes cleats to produce very subtle or elongated traces on borehole images, often appearing as faint vertical lines, not distinct sinusoidal curves. As a result, cleats produce elongated or ambiguous traces on borehole image logs, making it hard to distinguish from drilling-induced features or artifacts.

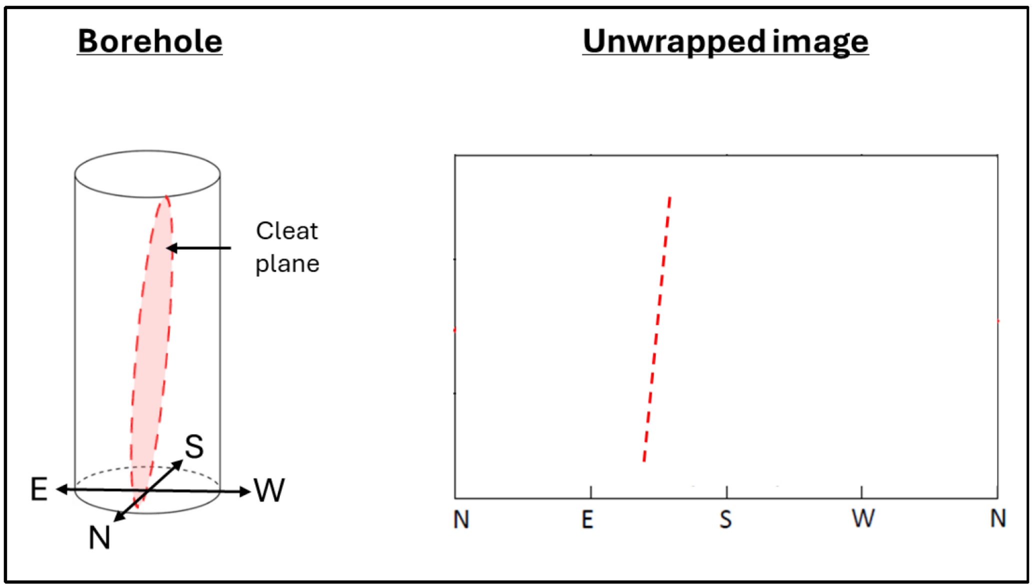

Figure 2—Cleat visualization in vertical boreholes, typically subvertical or steeply dipping. In borehole image log, the cleat is shown as a straight-line feature.

Despite significant advances in borehole imaging and analytical techniques, accurate cleat identification remains challenging especially when the image logs are captured from acoustic and optical scanners. The visibility of cleats on borehole images is low compared to other structural features such as bedding or fracture. Acoustic scanners rely on differences in acoustic impedance; cleats filled with similar material (e.g., mineralized or closed) may not generate strong reflections. Optical scanners need a clear visual contrast, but coal is dark and non-reflective, and the borehole fluid may be turbid, reducing image clarity. In vertical boreholes, vertical cleats offer very little lateral variation, making it hard to visually distinguish even with good imaging (Fig. 3).

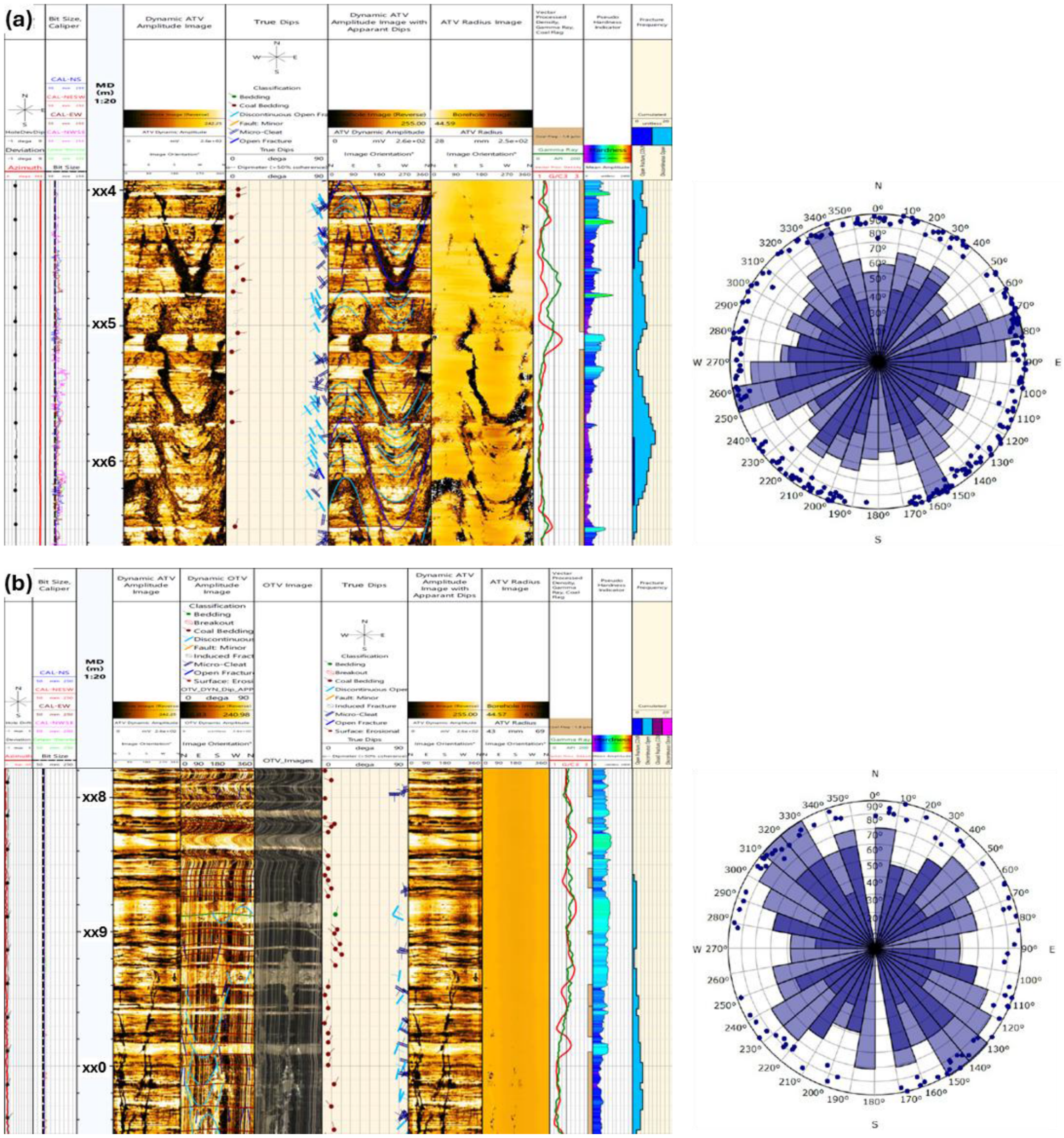

Figure 3—Cleat picking from different tools in vertical borehole (a) acoustic scanner and (b) optical scanner. Cleats were picked as a straight line parallel to the borehole. The orientation of cleats for both scanners are scattered, thus there will be high uncertainty in identifying face and butt cleat in vertical borehole.

Another concern of cleats picking in vertical boreholes is misinterpretation risk. Besides cleats, there are other vertical features that could be seen in borehole images such as bedding-parallel partings, drilling-induced fractures or tool artifacts. The visualization of cleats and other vertical features are quite similar, thus there is possibility of misinterpretation during the structural picking process. Besides, the orientation of cleats in vertical boreholes is generally scattered and it is difficult to identify face and butt cleats based on the stereonet (Fig. 3). Without additional data e.g. core or horizontal imaging, reliable cleat identification is uncertain.

Methodology

Borehole image plays a major role in structural interpretation applications for subsurface formation. By using specialized tool types such as acoustic, optical or microresistivity scanners, borehole images capture variations in any properties of the borehole wall. These images enable direct observation of structural features such as bedding planes, fractures, faults etc. For structural interpretation, borehole imaging facilitates the identification and orientation of planar and curvilinear features by recording their apparent dips and azimuths relative to the borehole axis. The resulting data can be processed to generate stereonet plots, rose diagrams, and fracture density maps, offering quantitative insights into subsurface structural frameworks.

Ten Water-Based Mud (WBM) horizontal boreholes were evaluated for the cleat identification along approximately 1 km NW-SE transect of a coal mining quarry (Fig. 4). Two boreholes are from single boreholes in two different locations and the remaining eight boreholes are dual-arm boreholes, which are from four locations. An 8-pad, 2.4" Slim Micro Resistivity Imager Tool (SMRIT) was deployed in 6" borehole to obtain high-resolution microresistivity image logs. Along with the image log data, petrophysical logs such as gamma ray, resistivity, density and caliper logs were obtained and integrated to provide a comprehensive understanding of the subsurface i.e. determine coal interval based on density logs.

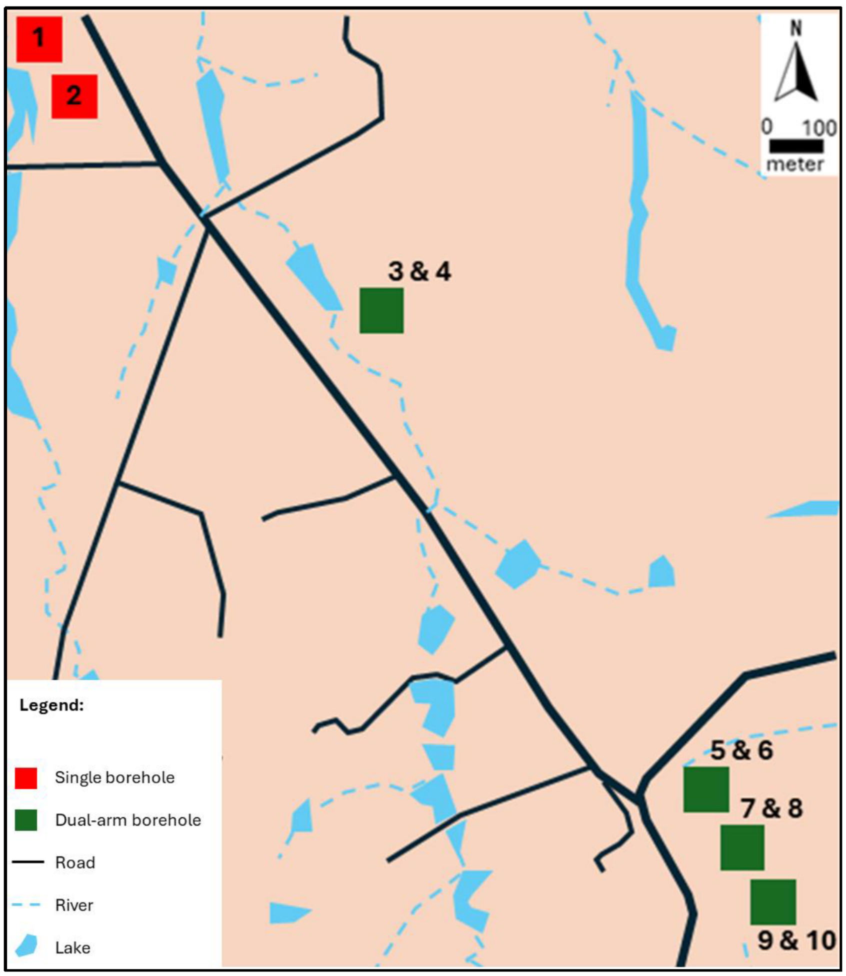

Figure 4—Distribution of the boreholes along the NW-SE section of study area; red dots represent single boreholes, and green dots represent dual-arm boreholes.

Further data processing was performed before producing static and dynamic images, including corrections for magnetic declination and speed correction. Magnetic declination was done to ensure the azimuth from magnetic north into true north coordinates. Speed correction was needed for irregular tool movement and cable and drill pipe stretch, which cause image log artifacts such as pad-to-pad mismatch and tool sticking. These corrections were crucial for generating both static and dynamic 360° resistivity-based borehole images.

Cleats were manually delineated from coal seams of each borehole, either in continuous or discontinuous sinusoid form. The dip direction of the cleats can be determined based on the trough of the sinusoid (Fig. 5) and the dip angles are more than 75°. Other structural features e.g. bedding and fractures were also picked for directional analysis. The number of cleats picked in each borehole were used to determine the density and spacing of cleats. The strike of the cleats was used to determine the major and minor direction of cleats based on the stereonet of each borehole.

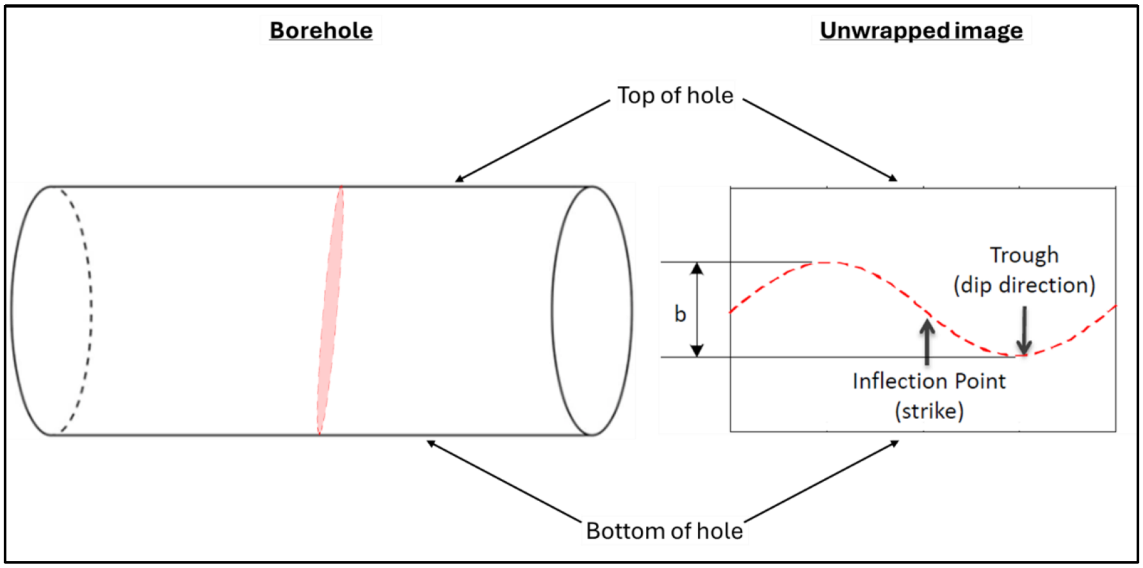

Figure 5—Identification of dip direction of cleat based on the trough of sinusoid form of cleat in the SMRIT image.

Structural Analysis of Cleats in Horizontal Boreholes

Cleat Identification

Based on the analysis of microresistivity borehole images acquired from 10 horizontal boreholes drilled through coal seams, cleats were clearly and consistently identified along the borehole trajectories using dynamic image displays (Fig. 6). These images provided a detailed and high-resolution view of the borehole wall, enabling the recognition of cleats that would otherwise be difficult to detect using other imaging methods. Unlike acoustic or optical borehole images, where cleat identification is often limited due to lower resolution or poor contrast in dark coal lithologies, SMRIT images allow cleats to be recognized in both continuous and discontinuous sinusoidal forms.

The presence of discontinuous cleats in the images is typically attributed to variations in image quality, such as pad coverage gaps or tool standoff issues. However, even when cleats appear discontinuous, they still retain valuable structural information. Notably, the trough of the sinusoid, which represents the point where the cleat intersects the lowest side of the borehole wall, can be used to determine the dip direction of the cleat. This makes discontinuous cleats reliable for directional or structural analysis, despite the visual interruptions.

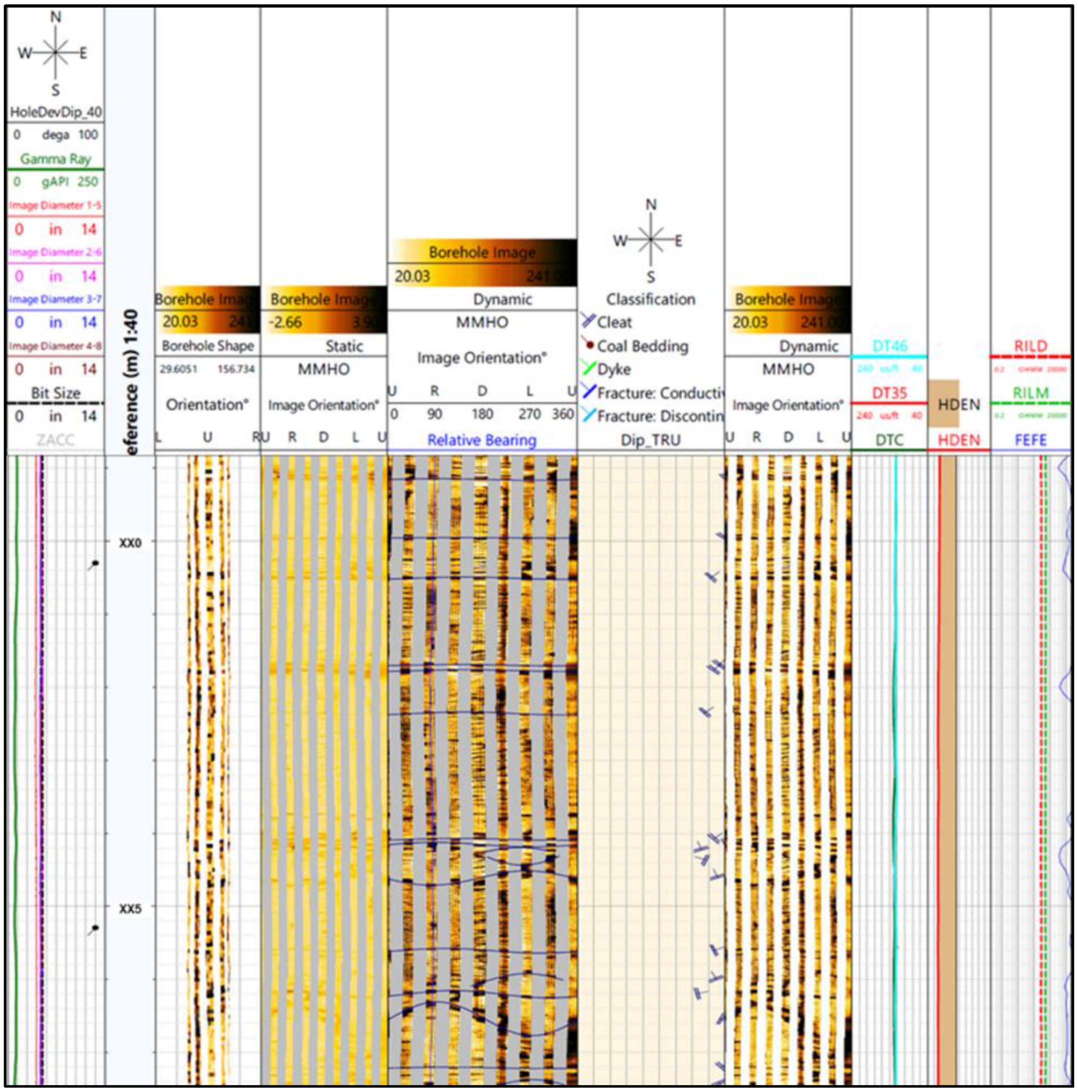

Figure 6—Cleat picking in SMRIT image log. Track 1 – auxiliary curves; Track 2 – measured depth; Track 3 – borehole shape plot; Track 4 – static SMRIT image; Track 5 – dynamic SMRIT image with dip picking; Track 6 – dip classification; Track 7 – dynamic SMRIT image without dip picking; Track 8 – compressional slowness; Track 9 – density log and Track 10 – resistivity logs. Cleats are clearly visible in dynamic SMRIT image. The visualization of cleats is either in full or discontinuous sinusoid form.

Furthermore, cleats can be effectively distinguished from other types of fractures such as conductive (open) or resistive (mineralized) fractures based on their dip angle. In this study, a dip angle threshold was applied to differentiate cleats from other features: cleats were defined as fractures with dip angles greater than 75° (Fig. 7). This distinction is important because cleats, being typically subvertical and orthogonal, show a consistent orientation and spacing that differ from randomly oriented or stress-induced fractures.

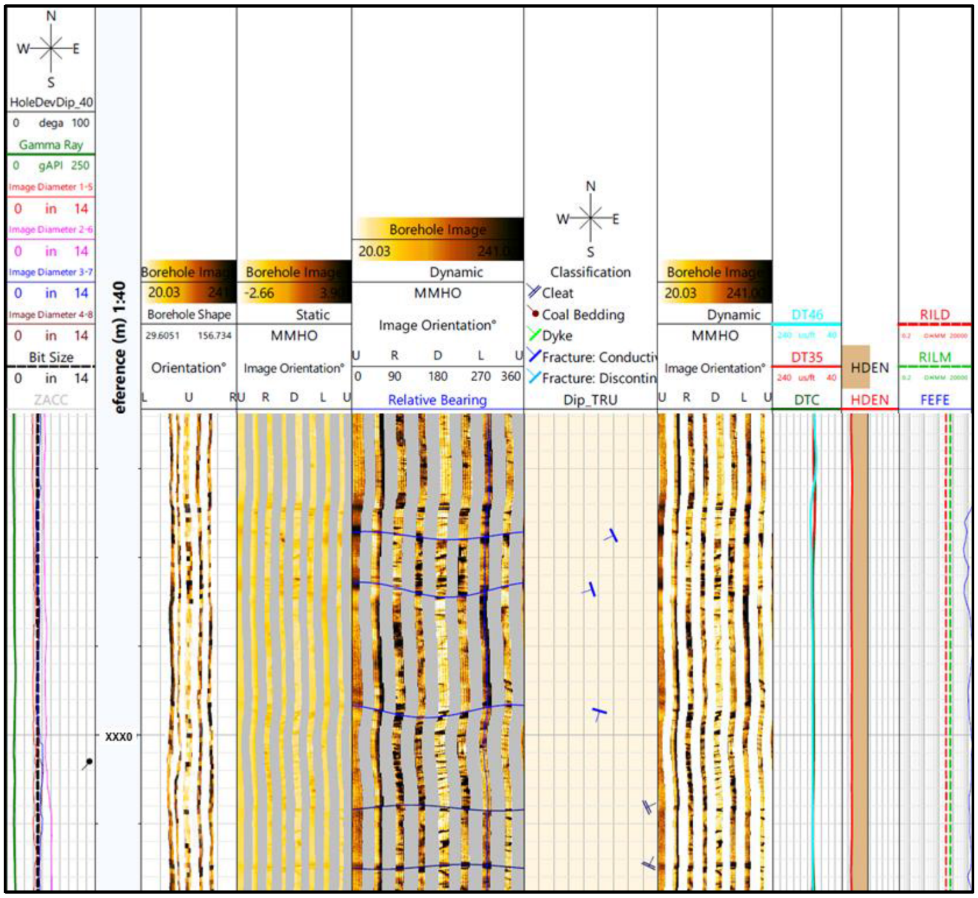

Figure 7—Difference between cleats and fractures in the horizontal borehole. Cleats will be determined based on a high dip angle which is more than 75°.

Cleat Density and Spacing

Based on the interpretation of cleats from borehole image logs, an average of 810 cleats were picked in each horizontal borehole, with each borehole having an average logged interval of approximately 500 meters. This translates to a significant presence of cleats distributed along the coal seam sections. In coal geology, a high density of cleats indicates the presence of a large number of systematically oriented, closely spaced natural fractures, comprising both face cleats (which are continuous and aligned with the principal stress direction) and butt cleats (which are shorter and typically intersect face cleats at near-right angles).

The cleat density in each borehole was calculated by dividing the total number of cleats identified by the length of the coal interval where cleats were picked. Among the boreholes analyzed, the dual-arm imager boreholes (Boreholes 5-10) exhibited a notably high cleat density, with values exceeding 1 cleat per meter (Table 1; Fig. 8). This high density of cleats indicates that cleats are consistently present at short intervals throughout these boreholes, suggesting enhanced fracture development and potentially better permeability in those coal seams.

In parallel, the cleat spacing, defined as the average distance between adjacent cleats (regardless of cleat type), was derived as the inverse of cleat density. As expected, boreholes with high cleat density correspond to low cleat spacing, which reflects a tighter fracture network (Fig. 8). Specifically, the cleat spacing values for Boreholes 5 to 10 are all less than 1 meter, confirming that cleats occur in tight and closely spaced arrangements within these intervals. Such tight spacing is geologically significant, as it may enhance the permeability and gas migration pathways within the coal seam, making these intervals of particular interest for CBM production or geomechanical evaluation.

Table 1—Cleats information obtained from cleat picking for each borehole.

Borehole	Depth Interval (m)	Cleats Picked	Cleats Density (m^{-1})	Cleat Spacing (m)	Major Direction	Minor Direction
1	x63	184	0.7	1.43	NW-SE	NE-SW
2	x88	291	0.75	1.33	NW-SE	ENE-WSW
3	x13	479	0.93	1.07	NW-SE	ENE-WSW
4	x90	455	0.93	1.07	NW-SE	NE-SW
5	x95	1148	1.44	0.69	NW-SE	NE-SW
6	x91	1019	1.47	0.68	NW-SE	NNE-SSW
7	x58	977	1.75	0.57	NW-SE	NE-SW
8	x40	2024	3.16	0.32	NW-SE	NE-SW
9	x11	580	1.87	0.54	NNW-SSE	E-W
10	x92	949	2.42	0.41	NNW-SSE	E-W

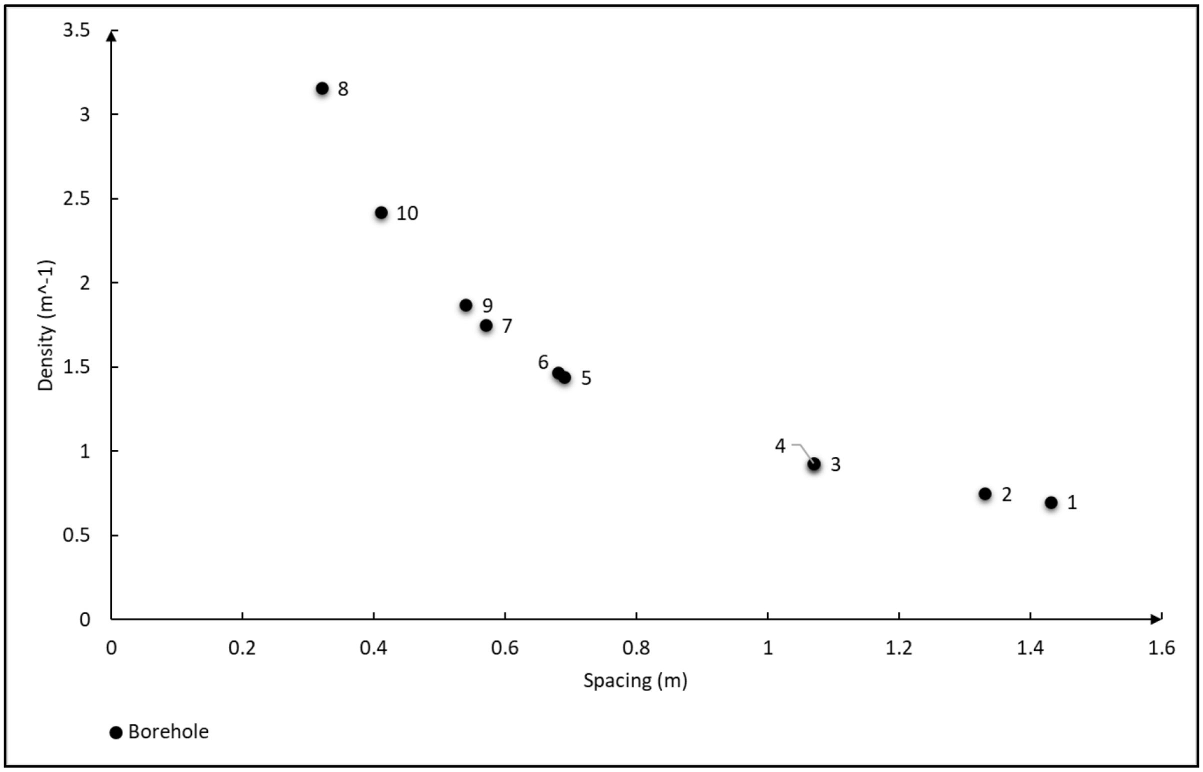

Figure 8—Relationship between density and spacing of cleats. Lower cleat spacing indicates higher cleat density.

A high cleat density within coal seams substantially increases the number of open, interconnected pathways available for fluid flow, thereby enhancing the effective permeability of coal, which is typically characterized by intrinsically low matrix permeability. This increase in permeability is particularly vital for CBM production, where the migration of methane gas and the efficient removal of formation water (dewatering) are both dependent on the cleat network. In such reservoirs, gas is primarily stored in an adsorbed state on the coal matrix, and its release and transport to the wellbore rely heavily on the extent and connectivity of cleats.

High-density cleat systems not only improve fluid flow but also influence the mechanical properties of the coal seam. The abundance of closely spaced natural fractures can weaken the structural integrity of the coal, potentially making it more susceptible to borehole collapse, instability during drilling, and challenges during completion operations. Moreover, these fracture systems can impact the effectiveness of hydraulic fracturing treatments by either enhancing fracture propagation through existing planes of weakness or by prematurely absorbing fracturing fluids, resulting in non-uniform fracture growth. In long-term production scenarios, high cleat densities may also contribute to subsidence or compaction of the coal seam due to stress redistribution and loss of pore pressure.

Furthermore, tight cleat spacing which indicative of high fracture frequency, translates into a more finely connected fracture network, which not only facilitates greater contact area for gas desorption but also promotes more efficient drainage of fluids toward the wellbore. This tight spacing directly improves the deliverability of both gas and water, accelerating production rates and enhancing overall reservoir performance.

However, while tight cleat spacing is beneficial for flow, it also means that the coal mass behaves as a more fractured and mechanically weaker medium, which can complicate drilling operations, affect wellbore stability, and alter the geomechanical response during stimulation treatments. Understanding the interplay between cleat density, cleat spacing, and coal mechanical behavior is therefore critical in designing effective CBM development strategies, optimizing well placement, and ensuring safe and efficient production over the life of the reservoir.

Cleat Directional Analysis

The orientation of cleats within the coal seams can be accurately interpreted using stereonet projections, which allow for the visualization of the spatial distribution and trends of fracture planes in three dimensions. By plotting the dip and dip direction of cleats identified from borehole image logs onto stereonets for each borehole, distinct directional patterns of cleat sets can be observed.

Based on the stereonet analysis conducted for each borehole, the major cleat direction, which corresponds to the face cleats, generally trends from NW–SE to NNW–SSE (Fig. 9). These face cleats are typically more continuous and better developed, and they often align with the maximum horizontal in-situ stress direction, reflecting their formation under regional stress regimes. In contrast, the minor cleat direction, representing the butt cleats, displays a trend that ranges from NNE–SSW to E–W (Fig. 9). Butt cleats are usually shorter and intersect the face cleats at high angles, often perpendicular or oblique to them, forming a systematic orthogonal fracture network typical of coal seams.

When focusing on the NW–SE transect of the study area, the dominant cleat orientation across multiple boreholes remains consistently aligned in the NW–SE direction, confirming a regionally persistent face cleat trend. Meanwhile, the butt cleats within this section show a subordinate but consistent NE–SW orientation, suggesting a structurally controlled cleat development influenced by local stress fields and coal fabric.

Mapping cleat orientation plays a vital role in determining the direction of σ_{Hmax}, which is a key parameter in the exploration and exploitation of CBM reservoirs. The orientation of in-situ stresses strongly influences the development, propagation, and openness of natural fractures within coal seams, particularly the cleat system, which consists of systematic face and butt cleats. These cleats form an orthogonal fracture network that provides the primary pathways for gas and water migration within low-permeability coal formations.

In coal geology, it is well established that the face cleat, which is the more continuous and hydraulically conductive fracture set, typically forms parallel to the direction of σ_{Hmax}, whereas the butt cleat, being shorter and often discontinuous, develops perpendicular to the face cleat and thus aligns with σ_{Hmin}. This orthogonal relationship reflects the influence of regional tectonic stresses during coalification and post-depositional deformation.

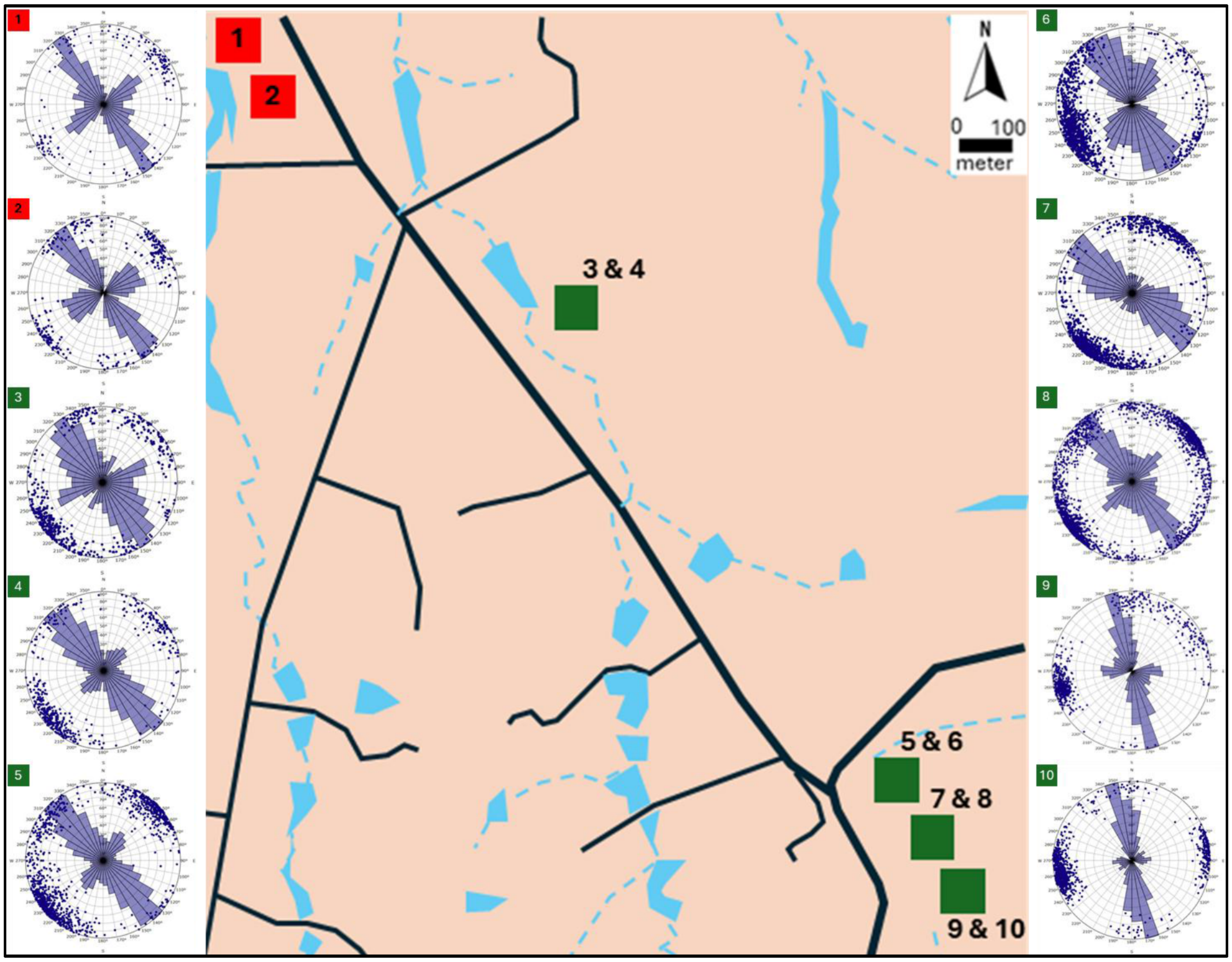

Figure 9—Stereonet of cleats in each borehole. Overall direction shows the major direction of cleats towards NW-SE direction while minor direction towards NE-SW direction.

Understanding the direction of σ_{Hmax} is essential because it controls the preferential orientation of fluid flow in both gas and formation water through the coal seam. In hydraulic fracturing operations, induced fractures tend to propagate in the direction of σ_{Hmax}, making the knowledge of stress orientation critical for designing effective fracture stimulation treatments, optimizing wellbore placement, and maximizing gas deliverability.

In the context of the current study area, cleat orientation analysis which derived from borehole image interpretation and stereonet projections, indicates that the face cleats predominantly trend NW–SE, suggesting that the direction of σ_{Hmax} is also NW–SE. Correspondingly, the butt cleats trend NE–SW, which aligns with the direction of σ_{Hmin}. This regional stress regime not only governs the natural fracture network but also directly affects reservoir permeability anisotropy, drainage efficiency, and reservoir management strategies for CBM development.

Conclusion

While acoustic and optical scanners are valuable borehole imaging tools, their ability to identify cleats in vertical boreholes is fundamentally limited by geometry, resolution, and imaging conditions. For reliable cleat analysis, horizontal or deviated wells with high-resolution imaging, supplemented with core data, are much more effective.

Based on cleats picking on horizontal boreholes using 8-pad SMRIT, we can reduce the uncertainty of cleat picking compared to vertical boreholes. The dip direction of cleats was determined accurately and could be plotted for further analyses. High density and tight spacing of cleats along the NW-SE section of study area suggests that the coals have high effective permeability despite it having low permeability. The face cleat of the study area shows the direction towards NW-SE direction and refers to σ_{Hmax}. While the butt cleat is trending NE-SW direction, represents σ_{Hmin}. This directional analysis could be integrated with geomechanical and laboratory analysis for coal seam reservoir development and field stress regimes.

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